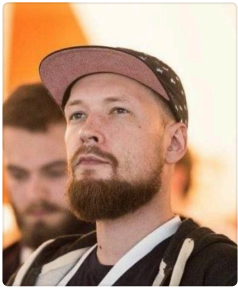


Sergey Rubanov

Building **AI agent infrastructure**, **web platform tooling & developer tools** — where specifications meet runtimes, engines, and developer-facing APIs.



✉ hello@sergey.works 🌐 sergey.works 🐙 chicoxyzy 🔗 chicoxyzy 🐦 sergey.works ✂ chicoxyzy

01 Profile

Software engineer and independent researcher working on AI agent infrastructure, the web platform, developer tools, and open source.

Works with **TC39** and **W3C** standards where specifications meet browser behavior, runtimes, and developer-facing APIs.

02 Core Skills

AI / LLM SYSTEMS

agent orchestration prompt/tool flows eval & quality-regression

harness llama.cpp MCP/ACP

WEB PLATFORM / STANDARDS

ECMA-262 TC39 Test262 WebAssembly JavaScriptCore WebKit

browsers/webviews

LANGUAGES

TypeScript Go C++ Zig

PLATFORM / OBSERVABILITY

OpenTelemetry traces/metrics/logs Prometheus Grafana

OTel Collector Docker Kubernetes Terraform AWS GCP

Cloudflare

03 Selected Projects

Hecate — AI RUNTIME TOOLING

Local-first console for local and cloud AI models, Hecate Chat, supervised coding-agent sessions, task approvals, usage visibility, and OpenTelemetry.

Stack Go · TypeScript · React · Tauri · SQLite · OpenTelemetry

🌐 hecate.sh . 🐙 github.com/hecatehq/hecate

Cynic — ECMASCRIPT ENGINE

Strict-only ECMAScript engine in Zig for non-browser hosts such as edge JS, Workers, and server-side JS.

Stack Zig · ECMAScript · Test262 · compiler/engine internals

🌐 sergey.works/cynic . 🐙 github.com/chicoxyzy/cynic

Pragmatist — VERIFICATION TOOLING

Tool for finding, reproducing, and verifying bugs across ECMA-262, TypeScript, and JavaScript engines.

Spec-aware queries, Coq/Rocq mechanization, TypeScript parser and library diffs, cross-engine differential runs, test262/spec patch scaffolds, and deterministic finding verification.

Stack TypeScript · Coq/Rocq · ECMA-262 · tsgo · Zod

Release coming soon.

TC39 MCP — MCP SERVER

MCP server that exposes TC39 specifications as clause-level, SHA-pinned data for agents and developer tools.

AOID-aware search, cross-spec references, edition diffs, git history, test262 and proposal lookup, offline stdio, and public HTTP.

Stack TypeScript · ECMA-262 · ECMA-402 · Cloudflare Workers · VitePress · Zod

🌐 mcp.xyzzylabs.ai/tc39/ . 🐙 github.com/xyzzylabs/tc39-mcp

Independent Software Engineer / Consultant

BARCELONA

Sep 2025 – Present

AI agent infrastructure, web platform standards, developer tools, observability, and open-source projects.

- Build AI agent and developer-tooling systems across tool execution, agent-task runtimes, web standards, JavaScript runtime internals, and observability.
- Design operator-facing reliability controls for agent systems: provider health/readiness, route diagnostics, trace links, usage/cost visibility, policy boundaries, approvals, and sandbox-aware execution.
- Contribute to TC39/ECMA-262 and Test262 by analyzing proposal trade-offs, refining specification text and semantics, and collaborating with browser vendors, JS engines, and the standards community.
- Work independently in ambiguous, research-heavy spaces: turn loosely defined agent/runtime ideas into usable tools, documentation, release paths, and test loops.

Invited Expert • TC39

May 2019 – Present

- Co-authored the **Promise.any** proposal and contributed editorial improvements to ECMA-262.
- Participate in proposal discussions, evaluate compatibility and ergonomics trade-offs, and develop Test262 tests to validate JavaScript behavior across engines.

Software Engineer • Firetiger

US • REMOTE

May 2024 – May 2025

Joined Firetiger after the acquisition of Dispatch.run; initially spent four months at Dispatch.run.

- **Dispatch.run**: owned the CLI for the Durable Functions platform, improving developer experience and local-to-cloud workflow reliability.
- Researched deployment approaches for running Go code through WebAssembly on Cloudflare.
- **Firetiger**: contributed to the telemetry product across core features, infrastructure, and OpenTelemetry compatibility.
- Supported early customers through onboarding, troubleshooting, integrations in Python/TypeScript, Vercel support, and Kubernetes/Terraform deployment workflows.

Stack Go • Python • TypeScript • Node • Next.js • Vercel • Docker • Kubernetes • Terraform • AWS / GCP / Cloudflare • OpenTelemetry / Prometheus / Grafana / OTel Collector

Software Engineer • Socket Supply Co

US • REMOTE

Mar 2022 – Jan 2024

- Developed core Socket Runtime components in C++ to improve execution stability and platform behavior.
- Enhanced JavaScript APIs for Node.js compatibility and native capabilities across mobile and desktop runtimes.
- Built tooling to streamline developer workflows and reduce setup friction.

Stack C++ / C / Objective-C++ • JavaScript / TypeScript • Node.js • WebKit- and Chromium-based webviews (WKWebView, libwebkit2gtk, WebView2)

Software Engineer • RealAdvisor

GENEVA • REMOTE

Dec 2019 – Sep 2021

- Developed a multi-tenant CRM and a custom document editor with layout templates to reduce manual formatting effort.
- Optimized the document rendering pipeline to improve load times and editor responsiveness.

Stack React / Svelte • Flow • Node.js • Relay • Docker + Composer • Headless Chromium • GCP

Frontend Developer Lead • EXANTE

MALTA • REMOTE

Dec 2014 – Aug 2019

- Led development of a web trading terminal, Chrome extension, Ledger hardware-wallet applications, and web3/cryptocurrency projects.
- Mentored and interviewed web developers while leading delivery across React/Preact, TypeScript, RxJS, D3, Redux/MobX, and browser-extension work.

Stack React/Preact • Redux/MobX • RxJS • D3 • TypeScript • C

Senior Developer • NPTV

MOSCOW • RUSSIA

Oct 2013 – Nov 2014

- Built a React-Native-like JavaScript SDK for an interactive TV platform; helped adopt SpiderMonkey and later migrate to V8.
- Managed build systems, DevTools extensions, and R&D for client applications.

Stack JavaScript • Babel • React • Chrome DevTools extensions

Frontend Team Lead · KM Media
MOSCOW · RUSSIA

Sep 2012 — Jul 2013

- Led frontend team to deliver the frontend for an advertising platform.
 - Established unit testing with Karma and Jasmine to reduce frontend defects and regression frequency.
 - Coordinated with backend and product teams to resolve integration issues and improve data accuracy.
- Stack** Angular · JavaScript

Senior Developer · Comfortel
ORYOL · RUSSIA

Sep 2009 — Jul 2013

- Built a CRM-like system with integrations for call-center software: Genesys Agent Desktop and Oktell.
 - Promoted to Team Lead in 2012; led a team to implement a CRM system, enhancing client interactions and driving measurable improvements.
 - Developed a dynamic web-based operator-script generator, streamlining call-center operations and reducing response times.
 - Managed and optimized databases (MySQL, MSSQL, Oracle), ensuring data integrity and system performance for critical applications.
- Stack** PHP · JavaScript · MSSQL · MySQL · OracleDB

Master of Computer Science · Oryol State Technical University
ORYOL · RUSSIA

2009

Languages — Russian (Native) · English (Proficient) · Spanish (Novice)